Solution Let R, r be the top and bottom radii of the frustum.

Let h_1, h_2 be the heights of the frustum and cylinder respectively.

Given that, $R = 12$ cm, $r = 6$ cm, $h_2 = 12$ cm

Now, $h_1 = 20 - 12 = 8$ cm

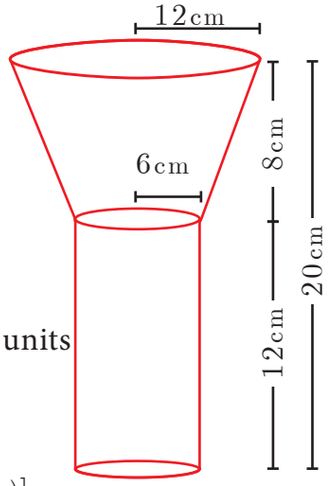


Fig. 7.41

Here, Slant height of the frustum $l = \sqrt{(R-r)^2 + h_1^2}$ units

$= \sqrt{36 + 64}$

$l = 10$ cm

Outer surface area $= 2\pi r h_2 + \pi(R + r)l$ sq. units

$= \pi\left[2rh_2 + (R + r)l\right]$

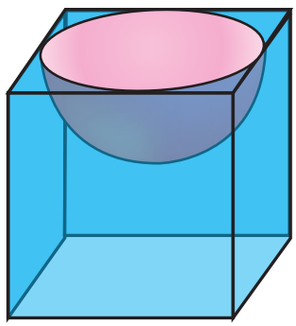
$= \pi\left[(2 \times 6 \times 12) + (18 \times 10)\right]$

$= \pi\left[144 + 180\right]$

$= \frac{22}{7} \times 324 = 1018.28$

Therefore, outer surface area of the funnel is 1018.28 cm^2

Example 7.28 A hemispherical section is cut out from one face of a cubical block (Fig.7.42) such that the diameter l of the hemisphere is equal to side length of the cube. Determine the surface area of the remaining solid.


Fig. 7.42

Solution Let r be the radius of the hemisphere.

Given that, diameter of the hemisphere = side of the cube = l

Radius of the hemisphere = $\frac{l}{2}$

TSA of the remaining solid = Surface area of the cubical part
+ C.S.A. of the hemispherical part
− Area of the base of the hemispherical part

$= 6 \times (Edge)^2 + 2\pi r^2 - \pi r^2$

$= 6 \times (Edge)^2 + \pi r^2$

$= 6 \times (l)^2 + \pi\left(\frac{l}{2}\right)^2 = \frac{1}{4}(24 + \pi)l^2$

Total surface area of the remaining solid $= \frac{1}{4}(24 + \pi)l^2$ sq. units

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